

Start mining CMFD

One beginner guide for Windows and Linux.
Includes terminal-only Linux and rented GPUs.

MAINNET LAUNCH EDITION

Downloads: October 2, 2026 at noon US Central (17:00 UTC).

Mining: October 3, 2026 at noon US Central (17:00 UTC).

These dates use CDT. Before publication, download links may return 404. Before activation, the miner waits. This guide does not start mining early.

This guide uses **pool mining**: your GPU submits work to a pool, and the pool handles full block proving. You do not need a desktop, wallet key, or full node on a rented mining server.

Before you begin

- An NVIDIA GPU with **at least 8 GB VRAM** and enough free GPU memory. The current search path uses about 6.45 GiB plus overhead; 6 GB cards do not fit. Capacity is not a guarantee that every card/driver is qualified.
- **At least 15 GB free disk for pool-miner setup**; 25 GB or more leaves useful headroom. A local wallet/node needs additional model and growing chain storage.
- Working internet, an accurate system clock, and a mainnet receiving address from a wallet you control.
- Permission to mine on the machine. For rentals, confirm provider policy, assigned GPU access and persistent storage before paying.

Your route through the guide

Everyone	Pages 2 and 7: downloads, wallet and success checks
Windows	Pages 3-4; verification on page 8
Linux / rented GPU	Pages 5-6; verification on page 9
Help and references	Page 10

Prepared September 28, 2026 from the signed v1.0.0 launchers. Mainnet-only: do not substitute older test-network packages or addresses.

SHARED STEPS / EVERYONE

Get the right files and address

1. Use the official release

Open [Common Foundry v1.0.0 downloads](#). On GitHub, expand **Assets**. Download the miner for the computer doing the mining. Also download **SHA256SUMS.txt** and **SHA256SUMS.txt.sig**. Do not choose GitHub's automatic "Source code" ZIP as the miner.

Windows miner

commonfoundry-mainnet-miner-windows-x86_64-v1.0.0.zip

Linux miner

commonfoundry-mainnet-miner-linux-x86_64-v1.0.0.tar.gz

Windows wallet + node

commonfoundry-mainnet-runtime-windows-x86_64-v1.0.0.zip

Linux wallet + node

commonfoundry-mainnet-runtime-linux-x86_64-v1.0.0.tar.gz

2. Verify before opening a program

Keep downloaded archives and the two checksum files in a dedicated downloads folder. Complete **page 8 for Windows** or **page 9 for Linux** before extracting/running the software. Those steps authenticate the signed checksum list and the archives you downloaded.

3. Create your wallet on a trusted personal computer

Download and verify that computer's **wallet + node** archive. Extract all its files. On Windows, open **START-WALLET.bat**. On a Linux desktop, run **./start-wallet.sh**. The Linux wallet needs a graphical GTK/WebKit environment; it is not an SSH-only wallet interface. The launcher may first download about 6.4 GB of authenticated model data.

Choose **Set up wallet**, then **Create new**. Choose a new backup-file save location, enter and confirm a strong passphrase (at least 12 characters), and create the encrypted wallet. Keep the encrypted backup somewhere separate and store the passphrase securely. There is no password-reset service that can recover an encrypted wallet for you.

Use **Show receiving address / Copy address / QR code** before launch, or **Receive** after connecting. The miner needs the **64-character hexadecimal public receiving address**, not a password, private key or backup file.

RENTED GPU? KEEP YOUR WALLET ELSEWHERE

Create the wallet on your own trusted desktop or laptop. Copy only its public receiving address to the rental. Do not upload wallet.key, encrypted backups, passphrases or signing keys to a rented server.

A GPU is not required for normal wallet/node verification. You may close the wallet after saving its address and backup; it does not need to stay open for a pool to pay that address. If you only have SSH access and no trusted wallet computer, ask official support before renting.

WINDOWS / PART 1

Make the GPU available to WSL

Use a supported Windows 11 installation with an NVIDIA GPU. The launcher is a Windows program, but its GPU worker runs in **WSL2 Ubuntu-22.04**. A Windows VM without GPU passthrough is not enough.

1. Install or update the NVIDIA Windows driver

Use the official NVIDIA driver for your GPU. Restart Windows if requested. Do not install a Linux NVIDIA display driver inside WSL. The package includes its CUDA runtime; a full developer CUDA toolkit is not required for these instructions.

2. Install the expected Linux environment

Right-click Start, choose **Terminal (Admin)** or **PowerShell (Admin)**, and run:

```
WINDOWS / Administrator PowerShell
wsl --install -d Ubuntu-22.04
wsl --update
```

Restart Windows if prompted. Open **Ubuntu 22.04** from Start and finish creating its Linux username/password. This Linux login password is not your Common Foundry wallet passphrase. Nothing appears while typing a Linux password; that is normal.

3. Check WSL2 and GPU visibility

In Windows PowerShell, run:

```
WINDOWS / PowerShell
wsl --list --verbose
wsl -d Ubuntu-22.04 -- nvidia-smi
```

The Ubuntu-22.04 row should show **VERSION 2**, and nvidia-smi should list your GPU. If it says version 1, run `wsl --set-version Ubuntu-22.04 2` and check again. Stop here if the GPU is not visible.

4. Install the small Linux prerequisites

Run these in the **Ubuntu 22.04 terminal**, not PowerShell:

```
UBUNTU / WSL terminal
sudo apt-get update
sudo apt-get install -y python3 curl ca-certificates libgomp1
```

If nvidia-smi is not on WSL's PATH, try: `wsl -d Ubuntu-22.04 -- /usr/lib/wsl/lib/nvidia-smi`. See [Microsoft WSL setup](#) and [NVIDIA WSL guidance](#).

WINDOWS / PART 2

Start your first miner

1. Extract the verified miner ZIP

Right-click the ZIP and choose **Extract All**. Use a simple writable location such as **C:\CMFD**, outside OneDrive and Program Files. Open the extracted folder that contains **START-MINER.bat**. Do not run it from inside the ZIP. Keep every packaged file and subfolder together.

2. Start with one GPU

Double-click **START-MINER.bat**. Paste your mainnet receiving address, then the complete mainnet pool URL when asked. Do not paste your wallet passphrase. GPU 0 is the default when no selector is set.

POOL CONNECTION FOR THE OPERATOR'S LAUNCH POOL

Use the following complete URL. It includes the port and certificate fingerprint. Confirm it with official launch instructions before use; a different pool needs that pool's own full pinned URL.

POOL URL / copy the entire single line

```
cmfd+tls://107.214.187.2:29445?pin=0a896a857857354781f06fe5b9ca05e632a35eab915b9cfd68545905bd1300d2
```

The pool is scheduled to become available on October 3 at noon Central. Before that, a launch-wait message is expected. Do not use an older network's pool endpoint just to make it connect.

3. Allow first-time preparation to finish

The miner downloads and verifies the model, obtains the authenticated launch certificate when available, then connects to the pool. This can take time. Keep the console open and watch for **MINER STATS** and an increasing **accepted** count (page 7).

4. Stop and restart safely

Press **Ctrl+C** to stop. To restart, open the same launcher again. Verified model data is reused; interrupted setup can resume. Keep Windows awake while mining. Closing its console or restarting Windows stops mining.

Optional: more than one GPU

List GPU indices with `nvidia-smi --query-gpu=index,uuid --format=csv,noheader`. For each assigned GPU, open a separate PowerShell window in the extracted miner folder. Replace 0 with that GPU's listed index:

WINDOWS / ordinary PowerShell

```
$env:CMFD_GPU="0"  
.\START-MINER.bat
```

One process per GPU. Use 1 in the second window only if GPU 1 exists and is yours to use. Do not start the same GPU twice. The miner adds a GPU suffix to the worker name. Wait for the first model download before adding more workers.

LINUX / PART 1

Prepare a terminal-only server

These commands target **Ubuntu 22.04 x86-64**, including an SSH-only GPU rental. No desktop or browser is needed on the mining server. Other distributions may need different packages; do not treat this as an all-distribution compatibility guarantee.

1. Connect and check your assigned GPU

Use the SSH command supplied by your rental provider. In the server terminal, run:

LINUX / SSH terminal

```
uname -m  
nvidia-smi  
df -h .
```

Expect **x86_64**, your assigned NVIDIA GPU, and at least **15 GB free disk** for setup. Confirm enough GPU memory is free. If `nvidia-smi` fails, ask the provider for a CUDA/NVIDIA-enabled image or GPU access; do not replace host drivers inside a rented container.

2. Install the required utilities

Use a normal account where possible. If the provider logs you in as root, omit `sudo`. If you cannot install software, ask for a suitable image.

LINUX / Ubuntu terminal

```
sudo apt-get update  
sudo apt-get install -y python3 curl ca-certificates \  
openssh-client tmux libgomp1
```

3. Download the official miner after publication

The commands below use a new folder under your home directory. On a rental, confirm that this folder is on persistent storage. If your provider requires another mount, substitute that location consistently.

LINUX / Bash

```
mkdir -p "$HOME/cmfd/downloads"  
cd "$HOME/cmfd/downloads"  
BASE="https://github.com/JustAResearcher/CommonFoundry-Binaries"  
BASE="$BASE/releases/download/v1.0.0"  
ARCHIVE="commonfoundry-mainnet-miner-linux-x86_64-v1.0.0.tar.gz"  
curl -fL --retry 3 -o "$ARCHIVE" "$BASE/$ARCHIVE"  
curl -fL --retry 3 -O "$BASE/SHA256SUMS.txt"  
curl -fL --retry 3 -O "$BASE/SHA256SUMS.txt.sig"
```

STOP AND VERIFY BEFORE EXTRACTING

Complete page 9 in this downloads folder. A 404 before October 2 is expected. Do not use an old release or a link sent in a private message as a substitute.

Budget rental time carefully: paying for a GPU before the network activates does not make it start mining early. The miner is separate from the wallet; your only wallet value on this server should be the public receiving address.

LINUX / PART 2

Mine and disconnect SSH safely

1. Extract the verified archive

Return to this step after the signature and archive checks on page 9 say OK.

LINUX / Bash

```
cd "$HOME/cmfd/downloads"
tar -xzf commonfoundry-mainnet-miner-linux-x86_64-v1.0.0.tar.gz \
-C "$HOME/cmfd"
cd "$HOME/cmfd/commonfoundry-mainnet-miner-linux-x86_64-v1.0.0"
```

2. Open a persistent terminal session

tmux keeps the miner session alive when your SSH connection closes. Create it once:

LINUX / Bash

```
tmux new -s cmfd
```

3. Set the public mining details and start

Inside tmux, paste this block. The launcher then asks for your **own 64-character mainnet receiving address**; paste it when prompted. If your assigned GPU has another index, replace 0 below with the index shown by `nvidia-smi`.

LINUX / inside tmux

```
export CMFD_WORKER_NAME="rental1"
export CMFD_GPU="0"
POOL_PIN="0a896a857857354781f06fe5b9ca05e632a35eab915b9cfd68545905bd1300d2"
export CMFD_POOL_URL="cmfd+tls://107.214.187.2:29445?pin=$POOL_PIN"
./start-miner.sh
```

This uses the operator's scheduled launch pool. Confirm its published pin before use. For another pool, use its complete officially supplied `cmfd+tls://IP:PORT?pin=...` URL instead. Follow the preparation, launch-wait and accepted-share checks on page 7.

4. Detach, return, or stop

Detach without stopping: press **Ctrl+B**, release both keys, then press **D**. You can now close SSH. Reconnect later and run:

LINUX / reconnect to your server

```
tmux attach -t cmfd
```

Stop mining: reattach and press **Ctrl+C**. To restart within that same shell, run `./start-miner.sh` again and re-enter your public address. After a server reboot, recreate the session and re-enter the settings. tmux does not survive a reboot or an expired rental.

For additional assigned GPUs, use separate tmux sessions such as `cmfd1` and repeat step 3 with GPU 1 and a distinct worker name. Never remove a provider's GPU isolation or use devices you did not rent. tmux reference: [official getting-started guide](#).

BOTH OPERATING SYSTEMS

Know when it is working

The normal startup sequence

- **Prepare:** download and authenticate the model. Pool mining needs the search model, not the roughly 61 GB solo-prover input set.
- **Wait if early:** the authentic launch certificate is required. Keep the clock correct; never modify the plan or substitute another beacon.
- **Connect:** the miner authenticates the pool using the certificate pin.
- **Submit work:** look for MINER STATS, a nonzero rate once initialized, and an accepted count that increases over time. A running process alone is not proof of accepted work.

The current miner reports **FW/s** (ForgeMatrix work per second), plus accepted/rejected/stale counters, temperature and power where available. Do not compare FW/s directly with another coin's H/s or MH/s. Rejected or stale shares can occur; a continuing high rate deserves investigation.

Common problems

Download says 404

Before October 2 noon Central, wait for publication. Afterward, check the exact v1.0.0 release and filename.

No GPU / CUDA unavailable

Check nvidia-smi. On Windows, check it inside Ubuntu-22.04 too. On a rental, ask the provider to fix assigned GPU access.

Out of memory

Stop only your own competing GPU apps, check free VRAM, and use an adequately sized card. An 8 GB label alone is not a compatibility guarantee.

Signature or checksum failed

Stop. Do not run the archive, disable checks, or accept a replacement key from a stranger. Re-download from official links and ask support.

Pool will not connect / pin mismatch

Check the launch time and the entire numeric-IP URL. Keep ?pin= and all 64 pin characters. Never bypass TLS verification.

Miner already running / GPU lock

Reattach to the existing terminal or tmux session. Stop your previous miner cleanly; do not launch duplicate copies or delete lock files.

Setup is slow / says it is waiting for a setup lock

Let the first instance finish. Check disk space and connectivity; rerun the same launcher after an interruption. Never use partial model files.

PAYMENTS ARE NOT INSTANT

The operator's launch pool is configured for a **0% operator fee**, a **10 CMFD payout minimum** and a **0.1 CMFD network burn per payout**. Rewards also need maturity (100 blocks). Block timing and share credit do not guarantee an immediate payout or profit.

VERIFICATION / WINDOWS

Check before you run the ZIP

Do this in **PowerShell** before running any downloaded program. Use a clean folder such as **C:\CMFD\downloads** containing only your chosen v1.0.0 ZIP archives, SHA256SUMS.txt and SHA256SUMS.txt.sig. If ssh-keygen is missing, install Windows' **OpenSSH Client** optional feature, not OpenSSH Server.

1. Obtain and check the independent public policy

WINDOWS / PowerShell

```
cd C:\CMFD\downloads
Invoke-WebRequest "https://commonfoundry.ai/mainnet-release-key.txt" `
  -OutFile release-key.txt
$trusted = "9c5be92681092801d89687823837823d0966055c03a84e64551e26387da179d2"
$actual = (Get-FileHash .\release-key.txt -Algorithm SHA256).Hash
if ($actual.ToLowerInvariant() -ne $trusted) {
  throw "STOP: release key policy does not match"
}
```

2. Authenticate the checksum list

WINDOWS / same PowerShell window

```
$signatureArgs = @("-Y", "verify", "-f", "release-key.txt",
  "-I", "commonfoundry-mainnet-owner",
  "-n", "commonfoundry-release", "-s", "SHA256SUMS.txt.sig")
$check = Start-Process ssh-keygen.exe -ArgumentList $signatureArgs `
  -RedirectStandardInput "SHA256SUMS.txt" `
  -Wait -NoNewWindow -PassThru
if ($check.ExitCode -ne 0) { throw "STOP: invalid signature" }
```

3. Check each downloaded ZIP against that signed list

WINDOWS / same PowerShell window

```
$zips = @(Get-ChildItem -File *.zip)
if ($zips.Count -eq 0) { throw "No ZIP archives found" }
foreach ($file in $zips) {
  $pattern = " " + [regex]::Escape($file.Name) + "$"
  $rows = @(Select-String -Path SHA256SUMS.txt -Pattern $pattern)
  if ($rows.Count -ne 1) { throw "STOP: archive not listed" }
  $hash = (Get-FileHash $file.FullName -Algorithm SHA256).Hash
  if ($hash.ToLowerInvariant() -ne $rows[0].Line.Substring(0,64)) {
    throw "STOP: archive checksum does not match"
  }
  Write-Host "$($file.Name): OK"
}
```

Continue only after the signature is good and every ZIP says OK. The public policy fingerprint is SHA256: cA1Ts f8hL / pxDV5WpOE3iPW3b4uDQosu1d0h / 4a / fk. These steps verify selected archives, not the complete 23-asset metadata reconciliation. See page 10 for the full release-verifier instructions.

VERIFICATION / LINUX

Check before you extract the miner

Run these in the downloads folder from page 5. Do not run any packaged program until the signature and checksum checks succeed. The key-policy file below is public; it is not a wallet or a private signing key.

1. Download the independent release policy

Use the official website, separately from the package download.

LINUX / Bash

```
cd "$HOME/cmfd/downloads"
curl -fL -o release-key.txt \
  https://commonfoundry.ai/mainnet-release-key.txt
TRUSTED="9c5be92681092801d89687823837823d0966055c03a84e64551e26387da179d2"
printf "%s  release-key.txt\n" "$TRUSTED" | sha256sum -c -
```

Expected result: **release-key.txt: OK**. If you do not see OK, stop. Do not change the expected policy hash to make it pass.

2. Verify the signature on the checksum list

This authenticates SHA256SUMS.txt under the release-signing namespace.

LINUX / Bash

```
ssh-keygen -Y verify -f release-key.txt \
  -I commonfoundry-mainnet-owner \
  -n commonfoundry-release \
  -s SHA256SUMS.txt.sig < SHA256SUMS.txt
```

Expected result: a **Good** signature for **commonfoundry-mainnet-owner**. If verification fails, do not continue.

3. Check the archive you downloaded

The `--ignore-missing` option lets you verify just the archive(s) you downloaded, rather than requiring all platforms.

LINUX / Bash

```
test -f commonfoundry-mainnet-miner-linux-x86_64-v1.0.0.tar.gz &&
sha256sum --check --ignore-missing SHA256SUMS.txt
```

The exact Linux miner archive must be listed with **OK**. If it is absent, reports FAILED, or the first command fails, stop. After success, return to page 6 to extract and run it.

WHAT THESE CHECKS DO - AND DO NOT DO

The signature authenticates the checksum list; the hash checks detect changed archive bytes. They do not prove an investment return, hardware compatibility, or absence of all software defects. The project's full verifier additionally checks the complete 23-asset set and build metadata.

Expected public-policy fingerprint:

SHA256:cA1TsF8hL/pxDV5WpOE3iPW3b4uDQosu1d0h//4a/fk

[HELP / SAFETY / REFERENCES](#)

Get help without risking your wallet

What to send to support

- Your operating system, GPU model, NVIDIA driver version and the exact v1.0.0 archive name.
- A screenshot or the last relevant error lines. Launcher logs are in **work/logs** inside the miner folder.
- Whether model preparation finished, the GPU is visible, and accepted shares are increasing. Say if you are on SSH, WSL or a rented container.

NEVER SHARE

Do not send private keys, wallet.key, encrypted backups, passwords, passphrase files, account tokens or remote-login credentials. Logs can contain your public wallet address and machine paths: review and redact them before posting. Support does not need control of your wallet to troubleshoot a miner.

Small glossary

Wallet address: Your public destination for payments. The miner expects 64 hexadecimal characters.

Worker name: A short label for a mining process, such as rental1. Use letters, numbers, dots, underscores or hyphens; maximum 32 characters.

Pool pin: A fingerprint that authenticates the pool connection. It is not a password.

Accepted share: Work the pool accepted. It is not the same as a confirmed block or a completed payout.

tmux: A Linux terminal session manager that can keep your miner running after SSH disconnects.

Official links and full verification

[Common Foundry website and official Discord entry point](#)

[Mainnet v1.0.0 downloads \(available from October 2\)](#)

[Full 23-asset signed release verification](#)

[Mainnet package user guide](#)

[Mainnet wallet backup and recovery](#)

[GPU qualification details](#)

The full release verifier requires all 23 release assets in a dedicated folder, Python 3.11+ and the pinned verifier dependencies from the published source. Keep this PDF, the independent key policy and extracted folders outside that exact asset folder. The Windows/Linux checks here intentionally cover selected archives only. Source-document links also become public at the scheduled source release.

Scope of this edition

Mainnet pool-mining instructions reviewed against signed source 033d41e45edf35033dfd4c9bf5f6f0774caf154b. This document is separate from the immutable signed release assets. It is not a test-network guide, a solo-mining guide, a physical test of every GPU, or a promise that the network is already live. Do not disable checks, use another network's address, or alter launch files to bypass waiting.

External setup references: [Microsoft WSL installation](#); [NVIDIA CUDA on WSL](#); [tmux getting started](#). Reviewed September 28, 2026.